Network Troubleshooting, Management, Monitoring, and Optimization

1. Use the Windows command prompt to run the command ipconfig /all on your computer and take a screenshot of the output. Identify and write down your IPv4 address, subnet mask, default gateway, and DNS servers as shown in the result.

IP Configuration Results (ipconfig /all)

Based on your attached screenshot, the network details for your Wireless LAN adapter Wi-Fi are:

Network Parameter	Value
IPv4 Address	192.168.1.23
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server (IPv4)	192.168.1.1
DNS Server (IPv6)	fe80::4663:c2ff:fe2a:15c8%15
DHCP Enabled	Yes
Network Adapter	MediaTek Wi-Fi 6 MT7921 Wireless LAN Card

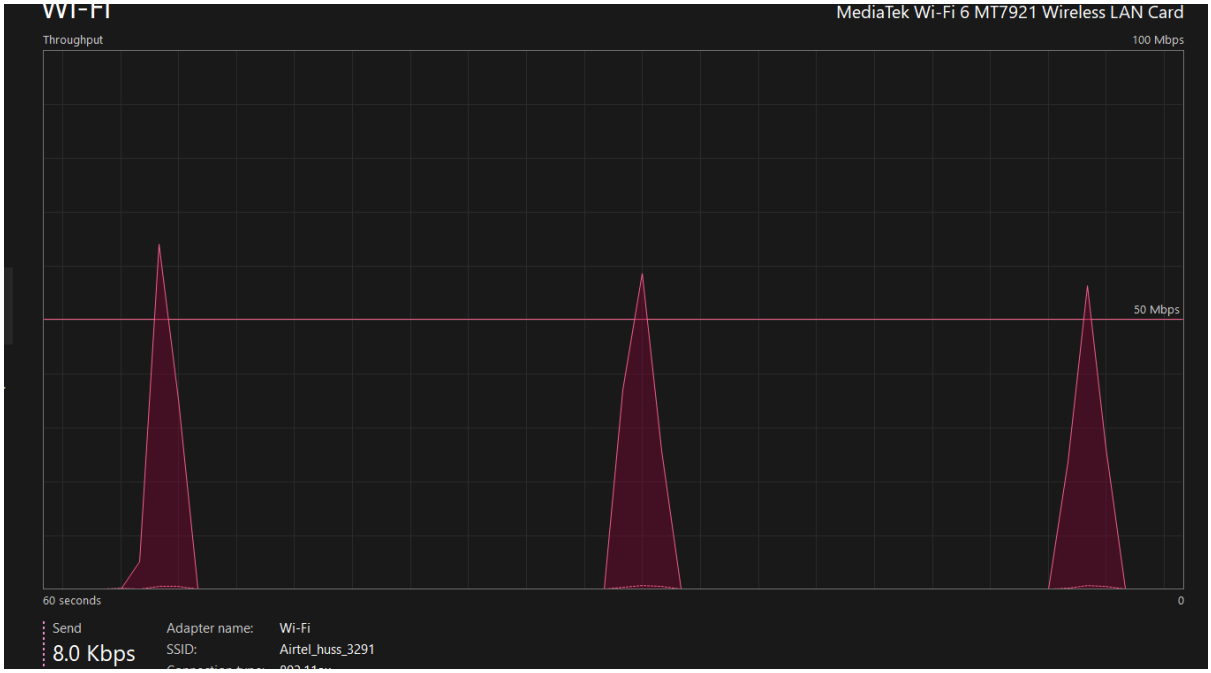
2. Simulate a network connectivity issue by disconnecting your Wi-Fi or Ethernet cable, then use the ping command to check connectivity to www.google.com. Reconnect and repeat the ping. Write a short note on the difference in results and what each status means.

Command used:

ping www.google.com

Network Status	Expected Result	Meaning
Wi-Fi disconnected	Ping request could not find host or General failure	The computer may be unable to resolve the hostname or send packets because the network is disconnected.
Wi-Fi reconnected	Reply from ...: time=XXms	The computer can reach Google's server and receives responses.
Successful ping	Sent = 4, Received = 4, Lost = 0 (0% loss)	All four packets received replies successfully.
Failed ping	Request timed out	No response arrived within the timeout; the cause could be a connection issue or blocked ICMP traffic.

3. Open Task Manager or Resource Monitor on your system and monitor network usage for 5 minutes while streaming a YouTube video. Record the peak network usage value and note which process used the most bandwidth during this time.



Network Usage Monitoring – YouTube Streaming

Observation: The Wi-Fi network activity was monitored using Windows Task Manager while streaming a YouTube video.

Parameter	**Observation**
Monitoring Tool	Windows Task Manager
Network Adapter	MediaTek Wi-Fi 6 MT7921 Wireless LAN Card
Wi-Fi Network	Airtel_huss_3291
Activity	YouTube video streaming
Peak Network Usage	Approximately 60 Mbps (estimated from the graph)
Process Using Most Bandwidth	Google Chrome

Result: The Wi-Fi throughput graph shows three significant spikes, with the highest reaching approximately 60 Mbps. Google Chrome is the expected process using the most bandwidth because it streams the YouTube video.

4. Download and install Wireshark (or use an online packet capture demo if installation is not possible). Capture at least 30 seconds of network traffic while browsing Flipkart or Instagram Web, then filter the packets to show only HTTP or HTTPS traffic. List 3 types of information you can see in these filtered packets.

Observation: Wireshark was used to capture and analyze network traffic while browsing websites. The screenshot shows encrypted HTTPS-related traffic using TLS 1.2 and QUIC protocols.

Findings:

1. **TLS 1.2:** Multiple packets are identified as TLSv1.2 Application Data, indicating encrypted communication between devices and remote servers.
2. **QUIC:** Some packets use the QUIC protocol, which supports fast, secure communication and is commonly used by modern web services.
3. **Source and Destination IPs:** The packet list displays IPv4 and IPv6 addresses, showing communication between the local network and remote servers.

Result: The Wireshark capture demonstrates that modern web traffic commonly uses encrypted protocols such as TLS and QUIC to protect data during transmission.

5. Imagine your friend is complaining about slow internet while gaming online. Use ChatGPT or any AI tool to generate a troubleshooting checklist (at least 5 points) for optimizing home network speed and reliability. Paste the AI's response and highlight one suggestion you would try first.

Troubleshooting Checklist: Slow Internet While Gaming

Here is an AI-generated checklist to improve home network speed and reliability during online gaming.

- ☐ Use an Ethernet cable: Connect the gaming PC or console directly to the router for a more stable connection and lower latency.
- ☐ Restart the router: Turn off the router, wait 30 seconds, and turn it back on to resolve temporary network issues.
- ☐ Reduce network congestion: Pause downloads, streaming, and other bandwidth-heavy activities on connected devices.
- ☐ Improve Wi-Fi signal: Place the router in a central, elevated location away from walls and electronic interference.
- ☐ Use the 5 GHz or 6 GHz Wi-Fi band: If supported, use these bands for faster connections at shorter distances.
- ☐ Check ping and packet loss: Run ping 8.8.8.8 -n 20 in Windows Command Prompt to check response times and packet loss.
- ☐ Update router firmware: Install available firmware updates to improve performance, stability, and security.
- ☐ Enable QoS (Quality of Service): If supported, prioritize gaming traffic to reduce competition from other devices.

Suggestion I Would Try First

I would first connect my gaming device directly to the router using an Ethernet cable. This can reduce Wi-Fi interference, improve connection stability, and help lower ping during online gaming.